
BUILDING, RUNNING & KILLING ALGOS

Jonas Löffler

loefflerjonas6@gmail.com

v1.0 · September 20, 2026

Abstract

A validated signal is not a strategy, and a running strategy is not a permanent one. This paper sits in the Trading Library's Algo-Development track: it takes a signal that has already survived honest backtesting and multiple-testing correction and carries it through specification, sizing, implementation, live operation, and — the part most treatments omit — disciplined retirement. Four claims organize the paper. Entry, exit, sizing, and risk are *one* specification, not four loosely coupled choices. Position sizing, not signal discovery, is where most real-world blow-ups are decided, so fractional Kelly and volatility targeting deserve at least as much care as the edge itself. An event-driven engine fed point-in-time data is the difference between a backtest and a look-ahead machine. And live operation is a measurement problem: reconcile live fills against the backtest, watch expectancy with a monitor whose alarm was set in advance, and know before you start what drawdown a healthy edge produces. The case studies are the author's own real algorithms — including strategies that were rejected, that flipped out-of-sample, and that were measured through an accidental bug — reported in aggregate, risk-normalized units only. A well-documented kill is worth more than three winners' curves.

Keywords — Strategy specification, position sizing, fractional Kelly, volatility targeting, risk of ruin, kill-switch, point-in-time data, look-ahead bias, live-versus-backtest reconciliation, slippage, CUSUM, drawdown bands, R-multiples, strategy retirement, portfolio construction, reproducibility

1 Introduction

1.1 Motivation and thesis

The literature on trading strategies is lopsided. An enormous amount is written about finding an edge and a fair amount about testing one, but very little about the two acts that consume most of a practitioner's actual time: turning a tested signal into a machine that trades unattended, and switching that machine off when its edge is gone. Those two acts — building and killing — are where money is actually made and lost, and they are governed less by cleverness than by discipline.

The thesis of this paper is that *the life-cycle of an algorithm is a single engineering and governance problem, and that the unglamorous ends of it — sizing, execution honesty, monitoring, and retirement — dominate the outcome*. A mediocre signal, sized sanely and retired promptly, is survivable. A brilliant signal sized to the edge you *hope* it has, implemented with a quiet look-ahead leak, and defended past its death because you are emotionally invested in it, is a route to ruin. This paper is written to keep the first outcome and prevent the second. It treats the algorithm not as a discovery to be celebrated but as a depreciating asset to be operated, measured, and eventually decommissioned.

Two ideas recur. The first is that *risk architecture is upstream of returns*: the sizing rule and the kill-switch determine the shape of the equity curve more than the entry signal does, and they are the only parts of the system fully under your control. The second is that *honesty is a technical property, not a virtue* — it lives in subscripts, in data timestamps, in whether the alarm threshold was set before or after you saw the drawdown. You do not become honest by resolving to be; you become honest by building the pipeline so that self-deception is expensive.

A word on why this paper leans so hard on the author's *own* failures rather than tidy textbook examples. The failure modes that matter are not the ones you can read about; they are the ones you commit, because they arrive disguised as good judgment — the profit target that "obviously" locks in gains, the filter that "clearly" improves things, the drawdown you are "sure" is different this time. A real bug that silently redefined a strategy, a real filter that reversed its sign the moment it met new data, teaches what no invented example can, because it happened to someone who was trying to be careful and was careful and still got it wrong. The point of Section 6 is not to display results; it is to show the reader the exact shape of the mistakes so they can recognize them in their own work before the market charges tuition for the lesson.

1.2 Scope, prerequisites, and intended reader

This is a research-grade paper in the Trading Library's Algo-Development track. It presumes three companion papers and does not re-derive them. It assumes *Backtesting & Hypothesis Testing*: pre-registration, the in-sample/out-of-sample split as a one-time event, expectancy and R-multiples, cost-sensitivity, and the direction check. It assumes *The Deflated Sharpe Ratio and Multiple Testing*: that the best of N trials is an order statistic and that the number of variants tried must deflate the significance of the winner. It also assumes *From Paper to Strategy*: the replication mindset and the paper-to-reality haircut. A reader without the backtesting and overfitting-control papers will mistake several arguments here for pedantry; they are the opposite.

Technically, the paper assumes comfort with a short Python snippet, a covariance matrix, and the language of Sharpe ratios, profit factors, and drawdowns. It assumes no stochastic calculus. It is written for someone who

already has a signal they trust and now has to decide, concretely, how many contracts to trade, what to do when the fills disagree with the backtest, and how they will know — in advance, with a rule — that the strategy has stopped working.

After reading it, a reader can:

- write a single, copyable strategy specification in which entry, exit, sizing, and risk are one coherent object, and defend every parameter as either load-bearing or removed;
- choose a position-sizing rule on purpose — fixed-fractional, volatility-targeted, or fractional Kelly — and state the growth-versus-drawdown trade-off in numbers rather than adjectives;
- build an event-driven backtest whose fills they trust and enumerate the code-level places where look-ahead hides;
- run a strategy live as a *measurement*, reconciling fills and slippage against the backtest and distinguishing a normal drawdown from a dead edge with a pre-committed monitor;
- set retirement criteria before going live and act on them without renegotiation; and
- combine surviving strategies into a portfolio whose risk-adjusted return beats any single component, and publish their own results without overfitting the narrative.

1.3 Data and reproducibility

The general figures (Section 3, Section 5, Section 8) are reproducible seeded simulations on no external data — each declares one fixed set-up in advance and reports whatever it produces, with no search over parameters or thresholds, precisely because the subject of this paper is the discipline that prevents such search. The one exception is the case-study figure in Section 6, which reads the author's private per-trade export; that list is **not redistributed**, only aggregate risk-normalized statistics are shown, and nothing in absolute currency appears. Values that came from a licensed or proprietary market-data feed are never reproduced here and are labelled *proprietary data source, not included here* wherever a signal depended on them. *This is a research and educational document, not investment advice.*

1.4 Organization of this paper

The paper follows the life-cycle in order. It is complete; every section below is present in full.

- **Section 2 From validated signal to strategy specification** — why entry, exit, sizing, and risk are one object; turning a signal into rules; parameter parsimony as a policy rather than a preference; a specification template to copy.
- **Section 3 Position sizing and risk architecture** — fixed-fractional, volatility-targeting, and Kelly sizing, and why fractional Kelly (Figure 1, full- versus half-Kelly statistics); portfolio-level risk, correlation, and kill-switches, with worked numbers.
- **Section 4 Implementation without lying to yourself** — event-driven engine essentials; point-in-time data; the code-level sources of look-ahead; paper-trading before capital.
- **Section 5 Live operation and monitoring** — live-versus-backtest reconciliation; slippage tracking; drift and health dashboards; what a *normal* drawdown looks like for a known edge (Figure 2, Monte-Carlo drawdown bands) versus the signature of a dead one (Figure 3, a rolling-expectancy and CUSUM monitor with a measured detection lag).

- **Section 6 Personal case studies** — four studies from the author's own work, reported in aggregate units, including the R-multiple distribution of the flagship case (Figure 4), several honest kills and rejections with post-mortems, and a family study in which the plain baseline wins. This section is flagged for the author's explicit publish review.
- **Section 7 Killing a strategy with discipline** — pre-committed retirement criteria; telling a regime pause from a death; the graveyard as an asset.
- **Section 8 A portfolio of strategies** — combining low-correlation edges; allocating capital across survivors; when the meta-portfolio beats any single algorithm (Figure 5, Sharpe comparison).
- **Section 9 Reproducibility and the ethics of publishing your own results** — what to show and what to redact; commit hashes; the line between honest self-marketing and overfitting your own narrative.
- **Section 10 Annotated references.**

2 From validated signal to strategy specification

A signal tells you *when the world is interesting*. A strategy tells you *what to do about it, with how much, and when to stop*. The gap between the two is where most of the risk lives, and it is routinely underestimated because the signal is the part that felt like insight. This section argues that the four components usually treated separately — entry, exit, sizing, and risk control — are a single specification, and that treating them as one is what makes a strategy testable, operable, and killable.

2.1 Four components, one object

Consider what happens when the components are designed in isolation. You find an entry signal with a positive edge. You bolt on a take-profit because "you should lock in gains." You add a stop because "you should limit losses." You size by a round number of contracts because that is what the account allows. Each decision is locally reasonable and the combination is untested — you have four independently plausible choices whose *interaction* determines the result, and you have measured none of them together.

The interactions are not subtle. A take-profit truncates the right tail of the trade distribution; if the edge lives in the right tail — as it does in every trend-following and breakout system — a "sensible" profit target quietly deletes the strategy's reason to exist. A stop interacts with the entry's natural adverse excursion: place it inside the noise and you convert winners into losers by getting shaken out; place it too wide and the position size implied by a fixed risk budget collapses. And sizing interacts with both: the number of contracts is a *function* of the stop distance whenever you size to a fixed risk per trade, so changing the exit silently changes the size. None of these can be evaluated one at a time. The specification is the object; the components are its coordinates.

This is not an abstract point. In the author's own equity-index work (Section 6), an intraday reversion signal was tested under six different exit structures on the same entries. The entry carried the edge in every case, but the exits moved net profit, drawdown, and — decisively — cost-sensitivity across a wide range, and the ranking of the six *reversed* once realistic costs were applied. The lesson is structural: you cannot rank exits, or choose a size, without holding the whole specification in view at once.

2.2 From signal to rules: making every decision codeable

A specification is finished when a competent programmer who has never spoken to you could implement it without asking a question. That test — *no ambiguity survives* — is demanding, and meeting it is most of the

work. Each of the following must become an exact, timestamped rule:

- **Entry.** The precise condition, evaluated on data available strictly at or before the decision bar, and the exact order type and reference price. "Buy the pullback" is not a rule; "if the fast signal is bullish and price touches the lower band, place a limit at the lower-band price for the next bar" is.
- **Exit.** Every way the position can end: target(s), stop, time-based flat, and any break-even or trailing logic — each with its own trigger and order type. If the position can scale out, the split is part of the rule, not an afterthought.
- **Sizing.** A function, not a number. Typically: $\text{contracts} = \text{floor}(\text{risk budget} \div (\text{stop distance} \times \text{point value}))$, bounded by a minimum and a maximum. The size is thus determined by the exit (the stop distance) and the risk policy jointly.
- **Risk control.** The conditions under which the strategy does *not* trade or trades smaller — session windows, regime gates, correlated-exposure limits, and the kill-switch (Section 3.5).

Two disciplines make this concrete. First, *specify the reference time of every input*. A rule that reads "yesterday's VIX close" is safe; a rule that reads "the VIX" is a bug waiting for a backtest to reward it, because in code "the VIX" too easily becomes the value at a bar you could not yet have seen. Second, *specify the order type and its fill assumption together*. A limit at the band and a market-on-touch are different strategies with different cost profiles; writing "enter at the band" hides the choice that a cost-sensitivity curve will later expose.

2.3 Parameter parsimony as a policy

Every free parameter is a small loan taken from your out-of-sample credibility, repaid with interest at the deflated-Sharpe counter (the overfitting-control paper). The expected best in-sample performance across the variants you try grows with the number of variants, so each additional knob you are willing to turn raises the bar the strategy must clear to be believed. Parsimony is therefore not an aesthetic preference; it is a pre-registered *policy* that caps the number of loans.

The policy has three rules. **One, prefer the specification with fewer parameters when two perform comparably** — not because simplicity is beautiful but because the simpler one borrowed less. In the author's VIX-gate work, a one-parameter gate (block short entries when the prior-day VIX is elevated) was preferred over a two-parameter version that added a second filter for a marginal net-profit gain; the second parameter did not pay for the credibility it cost. **Two, count the parameters you tried, not the ones you kept.** If you searched twenty threshold values and report the best, the strategy has twenty parameters for the purpose of significance, even though the specification shows one. **Three, freeze parameters at defensible values, not optimal ones.** A round threshold chosen for a mechanism ("elevated volatility begins around the low-20s in VIX terms") is more trustworthy than the argmax of an in-sample sweep, because the argmax is fitted to this sample's noise and the round number is not.

The uncomfortable corollary, developed honestly in Section 6, is that a striking result obtained by choosing the best of many variants on one dataset is *exactly* the kind this policy exists to distrust — including when the result is your own and you like it.

2.4 A specification template to copy

The following is a fill-in template for a single strategy. It is deliberately close to a pre-registration (the backtesting paper): the point is that the specification and its test plan are written together, before capital, and

frozen.

```
# Strategy specification – <id / short name>

## One-sentence thesis
- <the mechanism, with a direction: what inefficiency, why it should persist>

## Universe & session
- Instrument(s), resolution, trading hours, calendar exclusions.

## Entry (point-in-time)
- Condition: <exact, using only data known at/before the decision bar>
- Order: <type, reference price, next-bar timing>

## Exit (every path the trade can end)
- Target(s) / scale-out split: <levels + order types>
- Stop: <level + order type>
- Time stop / flat: <session time>
- Break-even / trail: <trigger + rule>

## Sizing (a function, not a number)
- Risk budget per trade: <in R or % of equity – never redistribute absolute $>
- contracts = floor(budget / (stop_distance * point_value)), min <...>, max <...>

## Risk & gates
- Regime / no-trade conditions: <...>
- Correlated-exposure cap across strategies: <...>
- Kill-switch: <pre-committed retirement criteria – see the pre-registration>

## Parameters
- Free parameters in the spec: <N_spec>
- Parameters/variants that will be TRIED in selection: <N_tried> -> deflate (the overfitting-control paper)

## Test plan (frozen)
- IS / OOS split (fixed dates), primary metric, cost grid, decision rule.

## Frozen on
- Date · commit · seed.
```

The two fields that most people skip are `N_tried` and the kill-switch. The first is what makes the eventual result believable; the second is what makes the strategy operable as a depreciating asset rather than a permanent article of faith. Both are developed in the sections that follow.

2.5 A worked signal-to-specification, end to end

To make the template concrete, here is the intraday reversion of Section 6 rendered from a one-line intuition into an unambiguous specification, with the interactions of Section 2.1 visible at each step.

The intuition: *"price tends to snap back to the session's average when it stretches too far, but not when the market is trending hard."* Rendered:

- **Universe and session.** One equity-index future, one-minute bars, regular cash hours only, with a hard flat before the close so no position is carried overnight (a different strategy, Case C, owns the overnight).
- **Entry.** Compute the session volume-weighted average price and its running standard-deviation bands. Require the average to be *sloping* in the trade's direction over a fixed look-back (the "not trending hard against us" clause), then place a limit at the first band — a limit, not a market order, because the thesis is

that price *reaches* the band and reverses, so you want to be filled *at* the stretch, not chasing it. The order rests for the next bar only.

- **Exit.** A stop beyond the second band (the point at which "stretched" becomes "trending," falsifying the trade); a break-even rule once price returns to the average; and a time-flat before the close. Whether to add a profit target is left *open* here precisely because Case A and Case D show it is the consequential, non-obvious choice — and one that must be decided by testing the whole specification, not by intuition.
- **Sizing.** Contracts = floor(risk budget ÷ (stop distance × point value)), floored at a minimum and capped at a maximum. Note the coupling the template forces into view: the stop sits beyond the *second* band, so in a wide-band (volatile) session the stop distance is large and the sizer buys *fewer* contracts automatically — the volatility targeting of Section 3.1, arriving for free out of the exit geometry.
- **Gates.** A regime gate on the prior-day volatility level (Case A's finding), a no-trade window at specified times, and a cool-down after a stop to avoid re-entering the same failing move.

Two things are worth noticing. First, every clause names its reference time and order type, so a stranger could code it. Second, the specification already contains the seeds of its own case studies: the "open" profit-target decision becomes Cases A and D, the sizing-exit coupling becomes the $\rho \approx 0$ finding of Case B, and the volatility gate becomes the multiple-testing caution of Case A. The specification is not just a plan to trade; it is a map of the decisions that will later have to be defended.

3 Position sizing and risk architecture

If the specification decides *what* you trade, sizing decides *whether you survive trading it*. This is the section where the largest, least recoverable mistakes are made, because sizing errors do not show up as a lower Sharpe — they show up as a zero. The core result is old and counter-intuitive: for a given edge there is an optimal bet size, betting *more* than it lowers long-run growth while raising risk, and the prudent choice is to bet a *fraction* of even the optimal amount.

3.1 Three sizing rules, in increasing sophistication

Fixed-fractional. Risk a constant fraction of current equity on each trade — operationally, set a risk budget equal to (say) a small percentage of equity and let the stop distance determine the contract count. Its virtue is that it scales exposure with the account automatically and can never, by construction, risk more than the budget on a single trade. Its cost is that it is blind to how volatile the opportunity is: the same budget buys wildly different real risk in calm and stormy markets unless the stop distance already encodes volatility.

Volatility targeting. Scale exposure inversely to a forecast of the instrument's volatility, so that the *portfolio's* risk, not the per-trade dollar risk, is roughly constant through time (López de Prado 2018). This is the workhorse of institutional systematic trading, and it has a subtle consequence that most retail treatments miss: *a stop-based fixed-risk sizer is already a volatility targeter*, because a wider stop in a volatile regime automatically buys fewer contracts. The author observed exactly this in a band-trailing intraday system: the correlation between the daily result and the VIX level was essentially zero (Spearman $\rho \approx -0.03$), and the reason was mechanical — the internal fixed-risk sizer had already flattened the volatility exposure, cutting the average contract count from roughly three in the calmest regime to barely over one in the most stressed. An additional volatility overlay on top of an already-vol-targeted sizer has almost no gross effect; its only real lever is net-of-cost, because trading many contracts in a low-volatility regime where the per-contract edge is thin is a way to pay costs for nothing.

The same point has a sharp opposite edge, and honesty requires stating it. Volatility targeting *assumes the edge is roughly constant across volatility regimes*. When it is not, mechanically scaling down in high volatility can be exactly wrong. In the author's overnight range-breakout system, the edge *rose* with volatility — the best nights were the most volatile ones — so the textbook "scale down when vol is high" rule would have thrown away the strategy's best opportunities. The rule is not "target volatility"; the rule is "sizing must match how the edge actually varies with the regime," and you must measure that rather than assume it.

Kelly. Given a known edge, the *Kelly criterion* prescribes the fraction of capital that maximizes the long-run growth rate of the equity (Kelly 1956; Thorp 2006). For a simple bet with win probability p , payoff b per unit staked on a win and the full stake lost on a loss, the growth-optimal fraction is $f^* = p - (1-p)/b$. Kelly is the theoretical ceiling: no other fixed fraction grows capital faster in the long run, and — the part that disciplines everything below — betting *more* than Kelly grows it *slower* while making the ride strictly worse.

3.2 Why fractional Kelly — the growth-versus-drawdown trade-off in numbers

Full Kelly is optimal only under assumptions that never hold in trading: that you *know* the edge exactly, that returns are stationary, and that you are indifferent to gut-wrenching interim drawdowns. In practice the edge is estimated with error, and betting full Kelly on an *overestimated* edge silently puts you past the growth-optimal point, where expected growth falls and drawdowns explode. The standard defense is to bet a fixed fraction of Kelly — commonly a half or less. Figure 1 quantifies why.

The set-up is fixed in advance: a hit rate of 0.45 and a payoff of 1.5-to-1, giving a per-unit edge of 0.125 and a Kelly fraction of about 8.3% of equity. One matrix of win/loss outcomes is generated once (seed 42) and reused across every staking fraction, so the curves differ only in sizing, not in luck. The left panel traces log-growth per trade against the multiple of Kelly; the right panel traces the median and 95th-percentile maximum drawdown over a thousand trades.

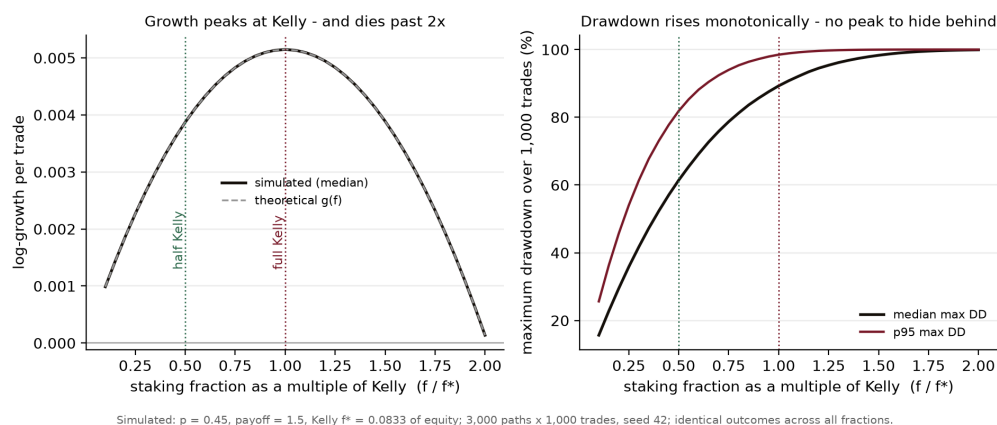


Figure 1. Growth and drawdown against the staking fraction, as a multiple of the Kelly bet, for a fixed edge (hit rate 0.45, payoff 1.5-to-1, Kelly fraction about 8.3% of equity). Left: median log-growth per trade peaks at full Kelly and turns negative beyond roughly twice Kelly. Right: the maximum drawdown rises monotonically with the fraction and has no peak to hide behind. One outcome matrix (3,000 paths of 1,000 trades, seed 42) is reused across all fractions, so the curves differ only in sizing. Reproduce with figures/fig_sizing_curves.py.

The numbers make the case that adjectives cannot. At **full Kelly**, the median terminal equity over a thousand trades is a multiple of about $\times 172$ — but the median maximum drawdown along the way is 89%, the 95th-percentile maximum drawdown is 98%, and 5.0% of paths end below where they started despite a genuine,

positive edge. At **half Kelly**, the median terminal multiple falls to about $\times 48$, yet the median maximum drawdown drops to 61%, the 95th-percentile drawdown to 82%, and only 0.8% of paths finish underwater. The crucial ratio: half Kelly retains about **75% of the full-Kelly growth rate while roughly halving the probability of an underwater outcome and taking tens of points off the drawdown**. Growth is quadratic and forgiving near the optimum; drawdown is monotone and unforgiving. You give up a little of a thing you cannot spend (interim growth rate) to buy a lot of a thing that ends careers (the depth of the hole). That is the entire argument for fractional Kelly, and it is why serious practitioners treat "full Kelly" as a boundary to stay well inside, not a target.

Two caveats keep this honest. First, the figure assumes the edge is *known*; with an estimated edge, the effective fraction you are betting is larger than you think, which pushes the prudent multiple below a half, not above it. Second, real strategies do not have i.i.d. binary outcomes — fat tails and serial dependence make the realized drawdowns worse than this clean simulation, again arguing for more caution rather than less.

3.3 From single-bet Kelly to a real strategy

The textbook Kelly formula is for one repeated bet; a strategy trades a distribution of R-multiples, often with a long right tail. The practical translation is not to compute a literal f^* — the inputs are too uncertain — but to *reason in Kelly's frame*: estimate the edge and its variance conservatively, note that the growth-optimal risk-per-trade is roughly the edge divided by the variance of returns, and then bet a fraction of that. For a strategy whose per-trade expectancy is, say, $+0.15R$ with a return standard deviation of a couple of R , the growth-optimal risk fraction is small, and a sane operating point is a small fraction of equity at risk per trade — precisely the fixed-fractional budget of Section 3.1, now justified rather than guessed. Kelly, in other words, is best used not as a calculator but as the reason the fixed-fractional budget should be *small*.

Case A's distribution (Figure 4) makes the danger of a naive Kelly calculation vivid. Its expectancy is $+0.165R$ and its variance is dominated by a handful of runner trades out beyond $40R$; plug those raw moments into a Kelly formula and the fat right tail inflates the variance so much that the "optimal" fraction looks tiny — but the tail is also where the profit lives, so the naive calculation both mis-estimates and misunderstands the strategy. The right lesson is not a number; it is that a heavy-tailed R-distribution is exactly the case where full Kelly is most treacherous, because the variance that Kelly divides by is itself estimated with enormous error from the few tail trades, and betting a fraction — a small one — is the only defensible response to that uncertainty. The heavier the tail, the smaller the Kelly fraction you should actually bet, and the less you should trust any single-number recommendation about it.

3.4 Portfolio-level risk and correlation

Per-trade sizing controls the risk of one position; it says nothing about the risk of holding several at once. The moment a book runs more than one strategy, the binding constraint becomes *joint* risk, and the governing quantity is correlation. Two strategies each risking a comfortable amount can, if they are correlated and fire together, risk an uncomfortable amount jointly — and correlations have the cruel habit of rising toward one in exactly the stress episodes when the joint loss matters most.

The architecture that follows from this has three layers. At the bottom, each strategy sizes its own trades to a fixed risk budget (Section 3.1). In the middle, a *correlated-exposure cap* limits the total risk that may be live across strategies that share a driver — for instance, capping the combined position when two equity-index strategies are both long, because in a crash they are one position wearing two names. At the top, a *total-heat* limit

bounds the sum of risk across the entire book regardless of correlation, so that a day on which many uncorrelated strategies happen to fire together cannot, by accident, put more than a pre-set fraction of capital at risk. These caps are not tuning parameters; they are the guardrails that make the difference between a bad week and a terminal one.

A worked illustration, in risk units to keep it dollar-free. Suppose each of five strategies is sized to risk $1R = 0.5\%$ of equity per trade, and suppose a total-heat cap of 3% of equity is set in advance. On a quiet day one or two strategies are live and total heat is well under the cap. But strategies cluster: a volatility spike can trip several equity-index strategies into the market on the same afternoon. If all five fire and all five happen to be positioned the same way, the naive risk is $5 \times 0.5\% = 2.5\%$ — under the cap, but the *effective* risk is worse than 2.5% because, correlated, the five are closer to one 2.5% bet than to five independent 0.5% bets, and their stops will tend to be hit together. The total-heat cap of 3% is the mechanical backstop that prevents a sixth or seventh correlated signal from compounding the day into something that a single bad hour could turn terminal; the correlated-exposure cap is the finer instrument that would have throttled the five *before* the total-heat limit was even approached, by recognizing they share a driver. The numbers are illustrative, but the structure is the point: per-trade sizing is necessary and not sufficient, and the two portfolio-level caps are what stand between an unlucky confluence and a career-ending day.

3.5 Kill-switches

A *kill-switch* is a pre-committed rule that halts a strategy — or the whole book — when a measured quantity crosses a line set in advance. It exists because the moment you most need to reduce risk is the moment you are least able to decide to. There are two kinds, and a serious operation has both. A *hard* kill-switch is mechanical and immediate: if the day's loss exceeds a set fraction of equity, or if the live system detects a data outage, a stale feed, or a fill it cannot reconcile, it flattens and stops trading with no human in the loop. A *soft* kill-switch is statistical and deliberate: the strategy's health monitor (Section 5) signals that the edge has plausibly decayed, and the pre-committed retirement criteria (Section 7) trigger a review that can only end in "continue on the documented evidence" or "retire." The hard switch protects you from the market and from your own infrastructure in seconds; the soft switch protects you from yourself over weeks. Both must be written down before the strategy goes live, because a kill-switch designed during a drawdown is not a kill-switch — it is a rationalization with a threshold attached.

The hard switch deserves the more paranoid design, because its failures are catastrophic and fast. A worthwhile set of triggers, all evaluated on the machine with no human veto: a daily-loss limit as a fraction of equity; a *data-staleness* trigger that flattens if the feed has not updated within a tolerance, because a strategy trading on a frozen quote is trading blind; a *reconciliation-break* trigger that halts if a live fill cannot be matched to an expected order within a window, because an unexplained fill means the system's model of its own state is wrong; and a *fat-finger* sanity bound that rejects any order more than a set multiple of the strategy's normal size. Each of these has, in real operations, been the difference between a bounded loss and an unbounded one, and each is trivial to code and easy to omit until the day it would have saved you. The design principle is that the hard switch should be *suspicious of the strategy itself* — it assumes the trading logic may be wrong and stands outside it, which is exactly why it cannot be part of the trading logic it is meant to catch.

4 Implementation without lying to yourself

Between a specification and a trustworthy result stands the backtest engine, and the engine is where honesty is won or lost in code. The backtesting paper established the discipline at the level of research design; this section is about the level below it — the software that computes the P&L — where a single wrong subscript manufactures an edge that was never there.

4.1 Event-driven essentials

The backtesting paper distinguished vectorized backtests, which compute signals and returns over whole arrays, from event-driven backtests, which step through the data bar by bar holding state. For a daily close-to-close study the vectorized form is safe and fast. But every strategy in this paper's case studies is *path-dependent* — it has intrabar stops, take-profit limits, break-even logic, and sizing that depends on the stop distance — and path-dependence is exactly the condition under which only an event-driven engine tells the truth.

An event-driven engine, at minimum, maintains explicit state for cash, open positions, and working orders, and it processes each bar in a fixed order that respects causality: first, *fill working orders* against the bar that could actually have filled them (a limit resting from the previous bar's decision, filled at the limit price only if the bar traded through it); then, *mark the position* and account the bar's P&L and any costs; then, and only then, *evaluate the signal on the just-closed bar* and place the orders that will be eligible to fill on the *next* one. The rule that prevents look-ahead is that decisions computed from a bar may only act on subsequent bars. An engine that computes a signal from a bar's close and fills at that same close has collapsed decision and outcome into one instant and will report a fictitious edge — the single most common way a backtest lies.

A discipline worth adopting from institutional practice: prototype vectorized to screen many ideas cheaply, then re-run every survivor event-driven, and *treat any discrepancy between the two as a bug to be explained, not a rounding difference to be ignored*. The discrepancy is almost always a look-ahead leak or an optimistic fill assumption that the fast version hid.

4.2 Point-in-time data

A backtest is only as honest as its data is *point-in-time* — meaning each value carries the timestamp at which it was actually knowable, not the timestamp it was later assigned. The failure modes are quiet and expensive. *Restated fundamentals*: an earnings figure revised months later, used at the original date, is knowledge from the future. *Survivorship*: an index membership list as it stands today, applied to the past, silently drops the companies that failed — the very companies whose omission flatters every long strategy. *Adjusted prices*: a back-adjusted futures continuation or a split-adjusted equity series encodes future events (the splits, the rolls) into past prices, which is fine for computing returns and ruinous for computing price-level signals.

The volatility complex, central to this paper's case studies, has its own version. A daily VIX value is stamped at the cash close; a strategy that decides intraday may use *yesterday's* close but not today's, and an analysis that merges the two must do so with an explicit as-of join that forbids exact-time matches, so that a signal computed at 09:36 never reads a value stamped 16:00 the same day. In the author's VIX-gate analyses this was enforced with an as-of merge that disallowed exact matches — the difference between a clean gate and a time-traveling one. The general rule: for every input, ask "at what instant did this number become knowable?" and make the code refuse to read it any earlier.

4.3 Code-level sources of look-ahead

Look-ahead is rarely a conceptual error; it is usually a line of code. A catalogue of the recurring ones, each worth grepping your own engine for:

- **The off-by-one fill.** Signal from bar t , position credited with bar t 's return. The fix is the subscript discipline of Section 4.1: exposure on t was decided on $t-1$.
- **The centered indicator.** Any smoother that uses future points — a centered moving average, a peak/trough labeled with hindsight, a "zig-zag" — leaks by construction. Only causal (trailing) transforms are admissible.
- **The full-sample fit.** Standardizing a feature by the mean and standard deviation of the *whole* series, or fitting a scaler on all data before splitting, leaks the future's distribution into the past. Statistics must be computed on a trailing window or fit on the in-sample block only.
- **The optimistic limit fill.** Assuming a limit order fills whenever price *touches* it, rather than trades *through* it, overstates fills on exactly the reversals a mean-reversion strategy depends on. The author's own family report flagged that its "fill at limit or better" assumption, without an intrabar magnifier, made the take-profit fills optimistic — a caveat that must ride with every number produced under it.
- **The silent repaint.** A signal that revises after the fact — an indicator that changes a past value when new data arrives — makes a backtest that can never be reproduced live. If a value can change after its bar closes, it is not a signal; it is a rumor.

4.4 Numerical determinism and the two-engine cross-check

A backtest that produces slightly different numbers on each run is not merely annoying; it is untrustworthy, because you can no longer tell whether a change in results came from a change you made or from the engine's own noise. Determinism is therefore a design requirement, not a nicety: every stochastic element — bootstrap resampling, Monte-Carlo paths, any tie-breaking that touches randomness — must draw from a seeded generator, and the seed, the library versions, and the data snapshot must be recorded with the result (Section 9.2). The general figures in this paper follow that rule: each caption names a seed, and re-running the script reproduces the printed numbers exactly. A result you cannot reproduce bit-for-bit is a result you cannot defend, and in a document that publishes its own performance that is disqualifying.

The strongest single safeguard against a silent look-ahead leak is to compute the same strategy two independent ways and require them to agree. The cheap, fast engine (vectorized) and the honest, slow one (event-driven) should produce identical P&L on any path-independent strategy; where the strategy is path-dependent, they should still agree once the event-driven engine's execution assumptions are matched to the vectorized approximation. A discrepancy is never "just numerical" — it is a specific, findable bug, and the direction of the discrepancy points at it: if the fast version is *better*, it is almost always fills the slow version correctly denies it (an optimistic touch-fill, an off-by-one entry), and the fast version is the one lying. Case A of Section 6 is the sharpest possible form of this lesson: there the disagreement was not between two of the author's engines but between the *written specification* and the *compiled strategy* — the profit-target ladder that the code described and the running system that never placed it — and it was caught only because the exit-tag tally, a form of two-source cross-check, showed zero fills where the specification demanded thousands. The general rule: never trust a single computation of your own edge; make two things that should agree, and investigate every time they do not.

4.5 Paper-trade before capital

The last gate before capital is a period of *paper trading*: running the finished system against the live feed, generating orders, and recording what would have filled — without money at risk. Its purpose is not to re-validate the edge (a few weeks cannot, and expecting it to is its own error) but to validate the *plumbing*: that the live data arrives on time and matches the backtest's, that orders are generated at the bars you expect, that the fill logic behaves, and that the monitoring and kill-switches actually fire. Paper trading is where you discover that the backtest used a data vendor's clean revised series while the live feed is messy and late, or that your "next-bar" order is really an "in-two-bars" order once real latency is included. It is a systems test, and the strategies that skip it tend to learn its lessons with real money instead.

5 Live operation and monitoring

Going live changes the epistemics of the whole enterprise. In the backtest you had a fixed dataset and unlimited time; live, you have a growing stream of the only truly out-of-sample data you will ever get — your own fills — and you have to make decisions before the sample is large enough to be sure. Live operation is therefore best understood as *measurement*: the strategy is a hypothesis, and every fill is a new observation that either is or is not consistent with it.

5.1 Live-versus-backtest reconciliation

The foundational live discipline, inherited directly from the reconciliation case in the backtesting paper, is to *run the backtest and the live system on the same data in parallel and reconcile them fill by fill*. For every live trade there is a backtest twin — the trade the model would have taken on the same bar — and the difference between them is the strategy's *implementation shortfall*: the sum of slippage, fees, latency, and any modeling error. Reconciliation answers the only question that matters in the first weeks live: *is the strategy underperforming because the edge is weaker than the backtest, or because the implementation is leaking?* Those two have opposite remedies — one is a reason to retire, the other a reason to fix execution — and you cannot tell them apart without the twin.

The reconciliation is also a bug detector of last resort. A persistent, one-signed gap between live and backtest fills — live always a little worse at the same price — is the fingerprint of an optimistic fill assumption (Section 4.3) that survived into production. A gap that appears only in fast markets is a latency or queue-position problem. A gap that appears only on certain instruments is a cost-model error. The discipline that in the backtesting paper turned a live shortfall into a corrected backtest is the same discipline that, run continuously, keeps a live strategy honest.

It helps to decompose the shortfall into named parts rather than a single lump, because each part has a different owner. For a given trade, the total gap between the backtest twin and the live result splits into *fees* (deterministic, known in advance, and the one part the backtest should already have modeled exactly), *spread* (the half-spread you cross, predictable from the instrument and knowable per fill), *slippage* (the movement between the decision price and the realized fill, the stochastic part Section 5.2 tracks as a distribution), and *modeling error* (everything left over — the residual after fees, spread, and slippage are accounted for, which *should* be near zero and, if it is not, is the signature of a bug in the backtest, not a cost of trading). The reason to split them is diagnostic: a shortfall that is all fees-and-spread means the strategy is behaving as modeled and the only question is whether the edge survives the honest cost; a shortfall with a large residual means the backtest and the live system are computing different things, and no amount of edge justifies trading a system you cannot reconcile to its own test.

The residual, in other words, is the number that decides whether you are looking at an economics problem or a software problem — and Case A of Section 6 is precisely a software problem that a fill-level reconciliation exposed.

5.2 Slippage tracking

Slippage deserves its own running measurement because it is both the largest controllable cost and the first thing to deteriorate as a strategy scales or as liquidity thins. The practice is to log, per fill, the *decision price* (the reference price the signal assumed) and the *realized fill price*, and to track their difference as a distribution over time, not a single average. The distribution matters because slippage is regime-dependent: it is benign in calm markets and vicious in the fast ones where a mean-reversion strategy does much of its trading, and an average over both hides the tail that actually hurts. When the measured slippage distribution starts drifting wider than the cost assumption baked into the backtest, the edge that the backtest promised is being eaten in production — a finding that must feed straight back into the cost-sensitivity curve of the backtesting paper and, if the edge dies at the newly measured cost, into the retirement decision of Section 7.

Two refinements make the measurement actionable. First, *separate slippage by order type*, because they have opposite signatures: a resting limit order, when it fills, fills at its price or better and its "slippage" is really adverse selection — it fills precisely when the market is about to move against it, so the cost hides in the fills you *did not* get as much as in the price you paid. A market or stop order, by contrast, pays visible slippage that widens with volatility and size. Reporting one blended slippage number over both order types averages two different phenomena into a figure that describes neither. Second, *track slippage against a size axis*, because the whole point of measuring it live is to find the capacity ceiling — the size at which the strategy's own orders begin to move the market against itself. A slippage that is flat across the sizes you have traded says you have headroom; a slippage that rises with size says you have found the strategy's capacity, and that ceiling belongs in the operating document next to the drawdown bands, because scaling past it converts a real edge into a donation to the order book.

5.3 What a normal drawdown looks like

The hardest judgment in live operation is also the most common: you are in a drawdown — is the strategy broken, or is this just what a working strategy feels like? The instinct to intervene is strongest precisely when intervention is least justified, because a genuine edge produces drawdowns that *feel* like failure and are not. The only defense is to compute, in advance, the drawdown distribution a *known* edge produces, so that a live drawdown can be read against a reference rather than against fear.

Figure 2 does this. A strategy with a real, unremarkable edge — a 40% hit rate paying +2R on wins and -1R on losses, an expectancy of +0.20R per trade — is simulated forty thousand times over four hundred trades (seed 42), and at every trade the cross-path quantiles of the worst drawdown seen so far are taken.

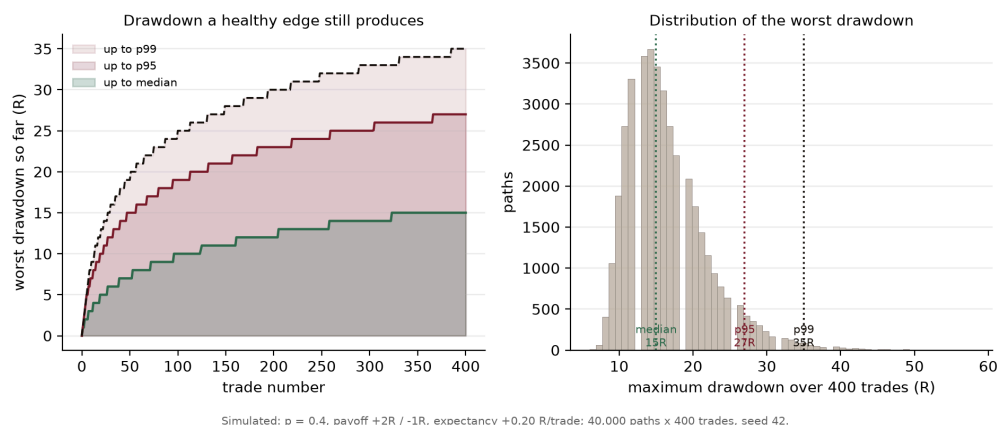


Figure 2. Monte-Carlo drawdown envelope for a genuinely positive edge (40% hit rate, +2R wins, -1R losses, expectancy +0.20R per trade), 40,000 simulated paths of 400 trades, seed 42. Left: the median, 95th- and 99th-percentile of the worst drawdown reached by each trade number, in R-units — the region a healthy edge still produces. Right: the distribution of each path's maximum drawdown over the full 400 trades. A live drawdown inside the p95 band is fully consistent with the known edge. Reproduce with `figures/fig_dd_bands.py`.

The numbers are sobering in the right way. For this ordinary +0.20R edge, the median worst drawdown over four hundred trades is about **15R**, the 95th-percentile worst drawdown is about **27R**, and the 99th-percentile reaches about **35R**. To put 27R in perspective: the *expected* terminal gain over four hundred trades is $400 \times 0.20 = 80R$, so a 95th-percentile drawdown of 27R is roughly a third of the total expected profit — endured, with 5% probability, by a strategy that is working exactly as designed. The operational consequence is direct: a drawdown of 20R in this strategy is not evidence of anything; it is inside the ordinary experience of a live edge. Only when the live drawdown pierces the p99 line — a level a healthy edge reaches just 1% of the time — does the drawdown itself become weak evidence that the edge has changed. Deciding this *before* going live, and writing the p95 and p99 levels into the strategy's operating document, converts the most emotionally loaded live decision into the reading of a pre-drawn line.

5.4 A health monitor for a decaying edge

Drawdown bands answer "is this drawdown normal?" They do not answer the subtler question: *has the edge quietly decayed even though the equity curve has not yet cratered?* For that you need a monitor that watches the expectancy itself and raises an alarm — on a threshold set in advance — when the mean result drifts down. Figure 3 builds one on a simulated strategy whose edge is engineered to die.

The strategy has a symmetric $\pm 1R$ payoff, so its edge lives entirely in the hit rate. For the first seven hundred trades the hit rate is 60% (expectancy +0.20R); then, over a hundred trades, it decays to 40%, flipping the edge to an equal-and-opposite -0.20R bleed. Two monitors run on the live R-stream, both fixed before the run: a hundred-trade rolling expectancy, and a one-sided lower CUSUM (cumulative-sum) on the standardized result. The CUSUM's reference is $k = 0.20\sigma$ (half the 0.41σ shift to be detected) and its alarm threshold is $h = 14\sigma$, chosen for an in-control average run length (ARL_0) of about **4,300 trades** — both set in advance, not tuned to the outcome. (An earlier revision quoted $ARL_0 \approx 3,300$, which is the Siegmund large-deviation approximation *without* the continuity correction; the exact in-control ARL_0 for this discrete CUSUM — from a collocation solution and confirmed by Monte Carlo at $\approx 4,290$ — is the right figure. The other monitoring metrics, median and p90 detection lag and the detection and false-alarm rates, are unaffected.)

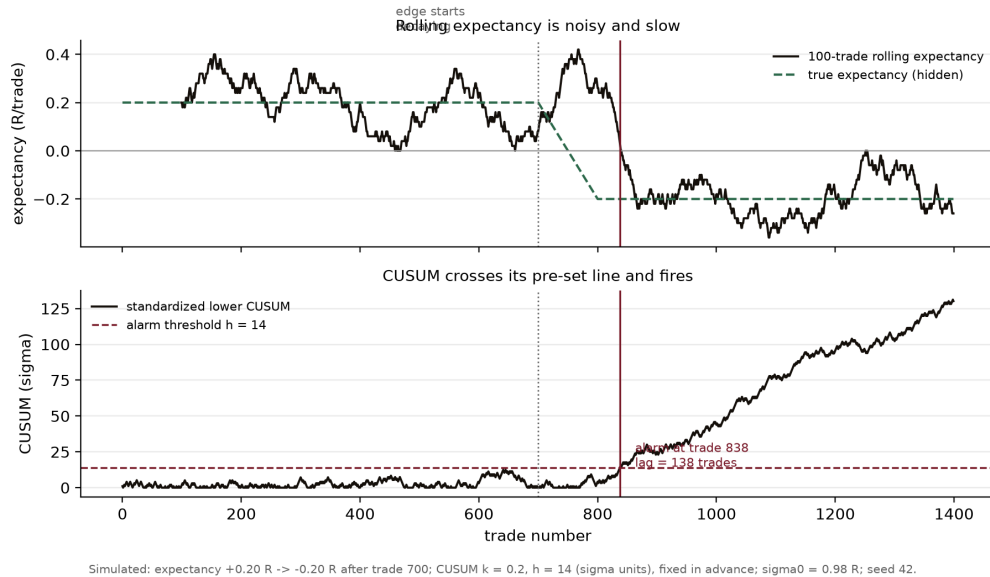


Figure 3. A health monitor on a simulated decaying edge. The strategy's hit rate falls from 60% to 40% starting at trade 700 (a symmetric $\pm 1R$ payoff, so expectancy drops from +0.20R to -0.20R). Top: the 100-trade rolling expectancy (noisy, slow) against the hidden true expectancy. Bottom: a one-sided lower CUSUM on the standardized result crosses its pre-set threshold and fires. Reference $k = 0.20\sigma$ and threshold $h = 14\sigma$ (in-control $ARL_0 \approx 4,300$ trades) are fixed in advance. On the illustrated path (seed 42) the alarm comes 138 trades after the changepoint. Reproduce with `figures/fig_monitoring.py`.

The result quantifies both the power and the limit of monitoring. On the illustrated path (seed 42) the CUSUM fires 138 trades after the edge began to decay; across five thousand independent paths the **median detection lag is 108 trades** (90th percentile 158), with the alarm firing correctly after the changepoint on about 85% of paths and raising a false alarm before it on about 15%. Two lessons follow. First, a well-designed monitor *does* catch a dying edge that the equity curve and the naked eye would miss for far longer — the rolling expectancy in the top panel is still a noisy blur when the CUSUM has already fired. Second, and more humbling, even a good monitor needs *roughly a hundred trades of a dead edge* before it is confident, because a per-trade edge is small next to per-trade noise. That lag is not a defect to be optimized away; it is the irreducible cost of distinguishing signal from noise, and it is the reason retirement criteria must be *pre-committed* (Section 7) rather than invented in the moment. You cannot make the monitor much faster without drowning in false alarms; what you can do is decide, before the money is on the line, what you will do when it fires.

6 Personal case studies

Publish-review flag. This entire section draws on the author's private, non-redistributed trade lists and on analyses that in places used a licensed market-data feed. Every number below is an *aggregate* — profit factor, Sharpe ratio, hit rate, R-multiple distribution, drawdown shape, or a train-versus-test ratio. No absolute capital, position size, per-contract cost, account value, or broker detail appears, and no value taken from a licensed feed is reproduced. **Jonas should review this section explicitly before publication** and confirm he can defend every figure in it. The purpose of the section is not to display winners; it is to show the discipline of this paper applied to real work, including where that work fell short.

These are four studies drawn from the author's own equity-index-futures work, over roughly four years of data (2022-09 to 2026-09). They are presented not as products but as specimens: each illustrates a specific lesson from

the sections above, and each includes an honest account of what went wrong. Three (A–C) are individual strategies; the fourth (D) is a family study that is really a lesson about testing exits, and about how often the honest answer is that the boring baseline wins.

6.1 Case A — an intraday reversion signal and its VIX gate

Hypothesis and build. The signal is an intraday mean-reversion on the E-mini S&P 500: fade a stretch away from a session volume-weighted average price when a slope filter agrees, entering with a limit at a standard-deviation band and sizing to a fixed risk budget. This is the flagship case, carried also through the backtesting paper's worked example. The variant studied here (internally "Idee 20") was designed to take profit at fixed R-multiples of the frozen stop distance — targets at 1.0R, 1.5R, and 2.5R across a three-contract ladder.

What actually happened — the bug that redefined the strategy. The first and most important finding is a post-mortem, not a result. Across the entire four-year backtest, *not one* of the designed R-multiple take-profit limits ever filled. The measured strategy was therefore not the one that was specified: with the profit ladder silently inert, what was actually being tested was a *no-take-profit trailing* strategy, whose gains came from break-even-band exits and end-of-day runners rather than from the intended R-targets. The generated exit code was logically correct; the compiled version behaved differently, and the discrepancy was caught only because the exit-signal tally showed zero fills at the ladder tags. This is a textbook illustration of Section 4: *the specification and the running system had diverged, and only a fill-level reconciliation revealed it*. The honest response was to relabel the result — "this is a no-TP strategy we measured by accident" — and to keep, from the accident, one genuine insight: the end-of-day runners were extraordinarily profitable on trend days, evidence that the right tail, not the intended targets, carried the edge.

Performance, in aggregate. Read as the no-TP strategy it actually was, the trade list has the characteristic shape of a low-hit-rate, fat-right-tail system. Figure 4 shows its R-multiple distribution, computed by normalizing each trade's P&L by the fixed per-trade risk budget the sizer targets — 1R by construction — since the export carries clean per-trade P&L but no per-trade stop distance and, as just noted, the R-ladder tags never fired.

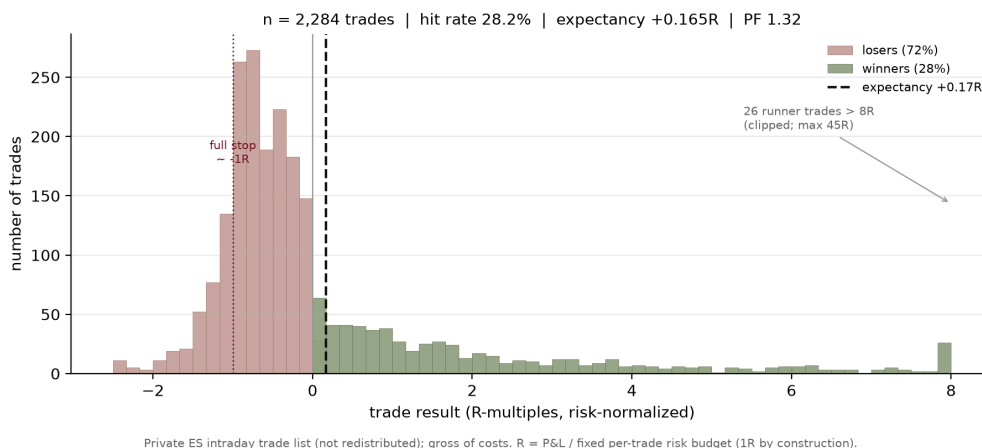


Figure 4. R-multiple distribution of the real intraday reversion trade list (Case A), 2,284 trades over roughly four years, gross of costs. R is each trade's P&L normalized by the fixed per-trade risk budget the sizer targets (1R by construction).

Losers cluster near a full $-1R$ stop and near zero (break-even exits); a thin right tail of end-of-day runners carries the expectancy. The dashed line marks the $+0.165R$ expectancy; 26 runner trades exceed $8R$ and are clipped at the right edge (largest about $45R$). Reproduce with `figures/fig_case_r_dist.py`. (private trade list, not redistributed)

The aggregate statistics: **2,284 trades, a hit rate of 28.2%, an expectancy of +0.165R per trade, and a gross profit factor of 1.32**. That profit factor of 1.32 is the same number carried into the backtesting paper's worked example, and it cross-checks exactly, which is the point of recomputing it here from the raw list. The distribution explains why a 28% hit rate is not a problem: the median trade is a small loss (about $-0.45R$, a break-even exit), and the strategy is profitable only because a thin right tail of runners — twenty-six trades beyond $8R$, the largest near $45R$ — pays for the many small losers. Truncating that tail with a profit target, as the *intended* specification would have done, might well have destroyed the edge; the bug, ironically, preserved it. Split at the pre-committed date, the out-of-sample expectancy ($+0.212R$) was actually higher than the in-sample ($+0.131R$) — a reassuring sign, but see the multiple-testing caveat below before reading too much into it.

The VIX gate — and why to distrust it. The most striking result in this case is a regime filter: blocking short entries when the prior-day VIX is elevated (below-20 only) raised the gross profit factor from **1.32 to 1.44**, cut the maximum drawdown, lifted the Sharpe ratio, and — the headline — *held out-of-sample, where the gated profit factor reached 1.62*. The mechanism is coherent: in volatility spikes, bear-market rallies run over the upper band and overrun short entries, while longs at the lower band are rewarded by panic rebounds, so short-side mean-reversion degrades monotonically as VIX rises while long-side reversion strengthens. The direction check passes, the split holds, and the story is mechanistically sensible.

And yet the overfitting-control paper and Section 2.3 demand that this result be distrusted, and honesty requires saying exactly why. The gate was *not pre-registered*. It was selected post-hoc from roughly twenty filter variants tried on the same trade list, and the threshold "20" was chosen in-sample. The in-sample/out-of-sample split shows directional stability, but it is not a walk-forward, and the high-volatility buckets that drive the effect come from only a handful of episodes — the effective sample is far smaller than the trade count suggests. By the standards this library holds itself to, the 1.44/1.62 result is therefore *formulated*, not *validated*: it is a promising candidate hypothesis generated by exploration, and its real test is a fresh pre-registered study on new data, with the twenty variants deflating the significance of the winner. This is precisely the case the prompt for a "kill" anticipated: absent a clean completed retirement, the most honest specimen is *a striking result that the multiple-testing guard requires us to hold at arm's length — including because we like it*.

The episode-clustering point deserves its own emphasis, because it is the caveat most easily lost behind an impressive trade count. The strategy's most extreme volatility buckets — the ones that most sharply distinguish the gated from the ungated result — contained on the order of two hundred trades, but those trades came from essentially three market episodes spread across the four years. A profit factor computed over "two hundred trades" sounds like a robust sample; a profit factor that in truth rests on *three episodes* is a sample of three, and one unusual month can move it substantially. This is the difference between the nominal sample size and the *effective* one, and it is exactly the illusion that intraday-frequency strategies invite: thousands of trades create an impression of statistical mass that a regime-conditional analysis can quietly undermine, because the regime of interest occurred only a few times. The honest reader of Case A should hold the 1.62 out-of-sample figure not as a precise measurement but as one draw from a distribution whose width is set by a handful of episodes — encouraging, worth a real pre-registered test, and nowhere near proof.

6.2 Case B — a band-trailing variant and the sub-signal that flipped

Hypothesis and build. The second case is a sibling of the first: the same intraday reversion entry, but with a band-trailing exit that uses the opposite band as a following stop ("Idee 14"). The research question, posed

explicitly, was whether the strategy's results correlate with the VIX regime such that one could size *up* when the regime is favorable.

What the data said — the sizing lever that was already pulled. The correlation between the daily result and the VIX level was essentially zero (Spearman $\rho \approx -0.03$ across 802 trading days), and this null is the interesting finding, not a disappointment. As developed in Section 3.1, the reason is mechanical: the fixed-risk sizer is already a volatility targeter, so the gross daily edge is flat across regimes by construction. The real, robust lever was net-of-cost economics, not gross timing — below a low-VIX threshold each contract's gross edge is thin enough that costs erase it, yet that is exactly the regime in which the tight bands make the sizer trade the *most* contracts. A regime-aware size tilt (cut risk in the cheapest regime, add a little in the richest) improved the net profit factor modestly, from about 1.25 to 1.28, and the Sharpe ratio from about 1.34 to 1.49, and under a conservative cost scenario it moved the strategy from net-negative to roughly break-even. The honest framing that accompanied the result in the vault is worth preserving verbatim in spirit: *a losing system does not become a winning system by tilting its size* — the overlay is a risk-smoother, not an edge.

The kill: a sub-signal that reversed sign out-of-sample. Within this case sits the cleanest falsification in the author's records. In the full-sample aggregate, a directional filter — "short entries lose once VIX is elevated" — looked convincing: the elevated-VIX short trades showed a negative average and a profit factor below one. But the pre-committed train/test split dismantled it. In the training window the elevated-VIX shorts averaged a *loss* ($n = 93$); in the test window the *same* filter's trades averaged a *gain* ($n = 59$) — a sign flip, not a weakening. A filter whose sign reverses out-of-sample is not a weak signal; it is a non-signal that happened to fit the training sample's particular episodes. It was rejected as a headline rule and retained, at most, as an optional and explicitly unstable component. This is the direction check from the backtesting paper operating at the level of a whole sub-strategy: the test-window evidence did not merely fail to confirm the filter, it *contradicted* it, and the correct response was to flag both the old belief and the new data and refuse to trade the rule.

A second kill in the same family. The band-trailing exit was also, straightforwardly, retired relative to its siblings. Compared across six exit structures on the same entries, it was the most cost-fragile — its winners were too small relative to its stops (an average take-profit-leg gain far below the average stop loss) — and under the conservative cost scenario it was the only variant to turn *net-negative*. The recommendation was explicit: discard it. No amount of VIX overlay repairs an exit whose payoff structure is upside-down; the lever that mattered was the exit itself, and the honest conclusion was to stop working on this branch.

6.3 Case C — an overnight breakout, and "leverage is not alpha"

Hypothesis and build. The third case is deliberately *uncorrelated* with the first two: an overnight range-breakout on the same index future, entering in the evening after a down day and exiting in the early hours — a different session, a different mechanism (overnight drift and mean-reversion after weakness rather than intraday band reversion). Its aggregate profile is a gross profit factor around 1.27, a hit rate near 57%, and a Sharpe around 0.8.

The finding that inverts the textbook. As noted in Section 3.1, this strategy's edge *rises* with volatility: the most volatile nights were the most profitable, and a backwardated term structure was the single sharpest condition for a good night. The correct sizing response is therefore the opposite of volatility targeting — scale *up* into the high-volatility regime, not down. A tiered scale-up improved the out-of-sample profit factor in the more conservative tiers (a floor tier moved from about 1.29 in-sample to 1.39 out-of-sample) while the most aggressive tier overfit and decayed out-of-sample (from about 1.76 to 1.53) — itself a small honest lesson about how far to push a regime tilt.

The rejection worth quoting. The most useful discipline in this case was terminological. A proposal to scale to two or three times the base size in favorable regimes *looked* like a risk-adjusted improvement — higher net profit, similar-looking ratios — until it was normalized to an equal drawdown budget, at which point the apparent improvement vanished. The finding was labeled precisely: *this is leverage, not alpha*. Scaling size up multiplies both the numerator and the denominator of a risk-adjusted return; unless the edge per unit of risk actually improves, a bigger number is just a bigger bet. Distinguishing the two — does this change the edge, or only the stake? — is the single most valuable habit in position sizing, and it is the habit most often skipped because a bigger equity curve is so persuasive. Several other overlays in this case were rejected on similar grounds: a hard high-volatility deadband that turned out to be fitted to the 2020 crash nights and was positive once those were excluded; and alternative volatility-forecast gates that were too collinear with the VIX to add anything.

6.4 Case D — the profit-taking family, where the null wins

Hypothesis and build. The fourth study is not a single strategy but an experiment about *exits*. Holding the intraday reversion entry fixed, six exit structures were tested on the same trades: the original full-target exit at a one-sigma band, and five profit-taking variants — a z-score-scaled target, a rejection-based exit, a progressive ratchet, the band-trail of Case B, and the R-multiple ladder of Case A. The question was the natural one after building an entry that works: *which exit makes it best?* It is exactly the question Section 2.1 warns is ill-posed when asked one component at a time, and the results show why.

What the data said — the boring baseline is hard to beat. Gross of costs, *no variant beat the original*. The plain full-target exit had the highest gross net profit of the six; the clever alternatives each traded some of it away. This is a result in the strict sense of the backtesting paper: the study's implicit null — that the elaborations do not improve on the baseline — was *not rejected*, and the honest report is the null itself. It is worth dwelling on how unusual it feels to write that, because the pull toward "surely one of five clever exits beats the naive one" is exactly the motivated reasoning that manufactures overfit strategies. Here the data declined to cooperate, and the correct output was a clean "no."

Costs reorder everything. The second finding is the one that changes practice. Applied gross, the six variants ranked one way; applied with two realistic cost scenarios, the ranking *reversed*, and it reversed differently under each scenario. The variants with many small partial fills — the ladders and trails — were punished hardest, because each partial leg pays the spread again; under the conservative scenario the band-trail of Case B went net-negative while a variant with fewer, larger winners rose to the top. This is the cost-sensitivity discipline from the backtesting paper at the level of a whole design choice: *the exit you would pick gross is not the exit you would pick net*, and a family report that stops at gross numbers has answered the wrong question. Report turnover and a cost grid next to every exit comparison, or do not compare them at all.

The multiple-testing coda. The third finding is a caution about the first two. Six exit variants were compared on the same four-year series, and the differences in their risk-adjusted returns were, over roughly four dozen months, statistically indistinguishable from noise. Six variants on one dataset is a small selection problem in its own right; before any of these differences could be acted on, the overfitting-control paper's machinery — a nested test against the baseline and a Sharpe deflated for the six trials — would have to be applied, and on this evidence it would very likely conclude that the family had discovered nothing beyond the entry it started with. That, too, is a result: the entry is the asset, the exit is a second-order tuning of it, and the honest end state of the whole family study is "keep the simplest exit that survives costs, and stop looking."

6.5 What the cases share

Read together, the cases make one argument four ways. In each, the *entry carried the edge and the interesting decisions were downstream* — in the exit structure, the sizing, and the regime handling. In each, the honest findings were the *negative* ones: a bug that redefined the strategy, a filter that flipped sign out-of-sample, an exit that died under realistic costs, a "risk-adjusted improvement" that was only leverage, and a family of clever exits that could not beat the plain one. And in each, the tools that produced those honest findings were the ones this paper is about: fill-level reconciliation, a pre-committed split, a cost-sensitivity curve, and the refusal to let a striking in-sample number wear the clothes of a confirmed one. None of the cases is presented as validated. That is not modesty; it is the correct status under the library's own rules, and stating it plainly is the whole point.

7 Killing a strategy with discipline

Every strategy dies. Edges are competed away, market microstructure changes, the regime that birthed a pattern passes, and the only question is whether you retire the strategy on your terms or the market retires it on its own. Retirement is the least-practiced discipline in trading because it runs against every incentive — sunk cost, identity, the hope that the last drawdown was the bottom — and so it must be governed by rules written when you were calm.

7.1 Pre-committed retirement criteria

The core discipline is to write the *retirement criteria before the strategy goes live*, as part of its specification (Section 2.4), and to make them measurable. A usable retirement clause names the quantity, the threshold, and the action, with no room for interpretation: *retire — or halve size pending review — if the live drawdown pierces the p99 band of the known-edge simulation (Section 5.3); or if the health monitor's CUSUM fires and stays fired for a further N trades (Section 5.4); or if the reconciled live expectancy is negative over a pre-set trade count; or if the measured slippage distribution shifts the cost-sensitivity break-even inside the realized edge*. Each of these is a line drawn on a chart you have already computed. The reason to pre-commit is not that your in-drawdown judgment is bad in general; it is that it is *predictably biased in one direction* — toward continuation — at exactly the moment the decision is live. A retirement rule is a contract with your calmer self, and its value is entirely in being unrenegotiable.

To make this operable rather than aspirational, a retirement clause should read like a contract, with the quantity, the threshold, the observation window, and the action all fixed. A worked example for the intraday reversion of Section 6, written in the strategy's own pre-committed units:

```
# Retirement clause – intraday reversion (frozen with the spec)

Halve size and open a formal review if ANY of:
R1 live max drawdown pierces the p99 known-edge band (~35R for this profile).
R2 the health-monitor CUSUM fires AND stays fired for a further 50 trades.
R3 reconciled live expectancy is <= 0 over any trailing 150 decided trades.
R4 measured slippage shifts the cost break-even inside the realized net edge.

Retire (flat, reallocate capital) if, at the review:
- the mechanism is judged broken (not merely the regime), OR
- a paused strategy fails a second review after its pre-set 6-month clock.

The review may ONLY conclude "continue on documented evidence" or "retire."
No new parameter may be added to rescue the strategy at a review.
```

The last two lines carry the weight. A review that is allowed to invent a new filter to explain away the failure is not a review; it is the moment a dead strategy is quietly re-fitted to its own drawdown, and it must be forbidden in advance because in the moment it will feel like insight.

7.2 Regime pause versus death

Not every failure is fatal, and the hardest call in retirement is distinguishing a strategy that is *dead* from one that is merely *paused* by an unfavorable regime it will survive. The two look identical in the equity curve and opposite in their remedy: a dead strategy should be retired and its capital reallocated; a paused one should be held, perhaps at reduced size, until its regime returns. Three questions discriminate them. *Is the mechanism intact?* If the economic reason the edge existed still holds and only the regime is wrong, pause is more likely than death; if the mechanism itself has been arbitrated or structurally changed, death is more likely. *Does the failure match a known regime dependence?* A strategy documented to struggle in, say, low-volatility trending markets, failing in exactly such a market, is behaving as characterized — a pause. *Is the failure inside the pre-computed bands?* A drawdown within the p95 envelope of Section 5.3 is not evidence of death at all, regime or otherwise. The synthesis is that death is a claim about the *mechanism* and pause is a claim about the *regime*, and you resolve them by asking which has changed. Crucially, "pause" is not a licence to hold forever — a paused strategy still needs a pre-committed clock, after which continued failure is reclassified as death.

7.3 The graveyard as an asset

A retired strategy is not a failure to be forgotten; it is a documented experiment whose record has ongoing value, and a serious operation keeps a *graveyard* — a log of every strategy it has run and killed, with the hypothesis, the performance, the retirement trigger, and the post-mortem. The graveyard pays three dividends. It prevents *rediscovery*: without it, the same seductive idea returns every eighteen months and is re-tested from scratch, its previous death forgotten. It *deflates future significance*: the graveyard is a running tally of how many things you have tried, which is exactly the N that the overfitting-control paper needs to correct the significance of your next winner — an operation that has quietly buried forty strategies should hold its forty-first to a far higher bar. And it is a *research asset*: the reasons strategies died — an exit that could not survive costs, a filter that flipped out-of-sample, an edge competed away as others found it — are themselves data about the market and about your own failure modes. The author's own vault functions this way: the rejected filters, the net-negative exit, the accidental bug, and the sign-flipped sub-signal of Section 6 are not embarrassments scrubbed from the record but entries in exactly this ledger, and their honest documentation is what allows the surviving strategies to be believed.

The deflation dividend is worth making precise, because it is the one most people's graveyards fail to deliver — not through malice but through amnesia. The overfitting-control paper's correction to the significance of your best strategy depends on N , the number of distinct configurations you actually tried, and the single most common way that correction is understated is that N is remembered as "the few I wrote up" rather than "everything I ran." A graveyard that logs only the strategies that reached live trading undercounts N by an order of magnitude, because for every strategy that went live, a dozen were killed in backtesting and forgotten. The discipline, therefore, is to log the deaths that happened *before* live as well as after — the variant that failed out-of-sample, the threshold sweep that found nothing, the exit family of Case D that could not beat its baseline — because those are trials too, and they belong in the N that will later be asked to deflate the next apparent discovery. A graveyard kept this way turns the demoralizing fact that most ideas fail into a quantitative asset: the failures are the denominator that makes the rare success statistically credible.

8 A portfolio of strategies

A single strategy, however good, has a single point of failure and a single drawdown to sit through. The strongest argument in systematic trading is not for a better strategy but for a *portfolio* of merely adequate ones whose edges are weakly correlated — because diversification improves risk-adjusted return in a way that better signal selection, past a point, cannot.

8.1 Why low correlation beats high Sharpe

The arithmetic is the whole argument. Combine several strategies with equal-risk weights; the portfolio's return is the average of theirs, but its volatility is *less* than the average of theirs whenever they are less than perfectly correlated, because the losses of one partly cancel the gains of another. The portfolio Sharpe ratio therefore exceeds the average component Sharpe, and the improvement grows as correlation falls.

A clean special case makes the magnitude tangible. Take n strategies with the *same* Sharpe s and the *same* pairwise correlation ρ , combined with equal weights. The portfolio Sharpe is $s \times \sqrt{n / (1 + (n-1)\rho)}$. With $\rho = 0$ the bracket is $\sqrt{n}$: three independent equal edges combine to $\sqrt{3} \approx 1.73$ times the single Sharpe — a strategy of Sharpe 0.9 becomes a book of Sharpe 1.56 with no better signal. As ρ rises the benefit erodes: at $\rho = 0.15$ the multiplier for three strategies falls to $\sqrt{(3 / 1.30)} \approx 1.52$, and as $\rho \rightarrow 1$ it collapses to 1, because perfectly correlated strategies are one strategy and diversification buys nothing. The entire value of a second strategy is in the gap between its correlation to the first and one, which is why the search for a *genuinely different* edge dominates the search for a slightly better version of an edge you already own. Figure 5 makes this concrete with three simulated edges deliberately chosen to resemble the author's three real families — an intraday reversion, an overnight breakout, and a volatility mean-reversion — with standalone Sharpe ratios of 0.90, 0.80, and 1.00 and low pairwise correlations (0.10 to 0.20). Combined with equal-risk (inverse-volatility) weights, the population Sharpe ratios are exact rather than a noisy single draw:

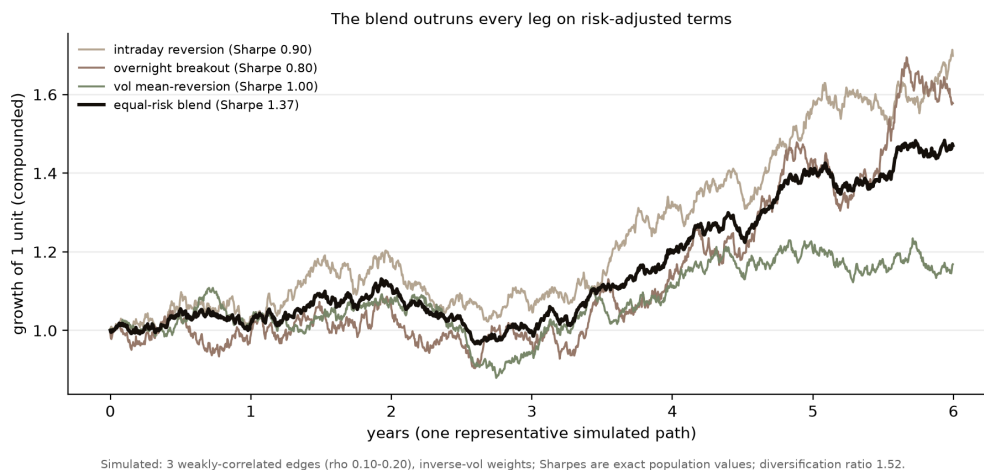


Figure 5. Three weakly-correlated simulated edges (standalone annualized Sharpe 0.90, 0.80, 1.00; pairwise correlations 0.10 to 0.20) combined with inverse-volatility weights. The equal-risk blend has an exact population Sharpe of 1.37 — higher than any single leg — with a diversification ratio of 1.52. The curves are one representative six-year path; the legend Sharpes are the exact population values, not that single draw. Reproduce with `figures/fig_meta_portfolio.py`.

The blend's Sharpe ratio is **1.37** — **above the best single leg's 1.00**, an uplift of +0.37 achieved with no better signal, only diversification, at a diversification ratio of 1.52. That uplift is *free* in the specific sense that it

required no improvement in any component; it came entirely from the low correlations. This is why the marginal hour is better spent finding a genuinely different edge than squeezing a few more basis points out of an existing one: a fourth weakly-correlated strategy of even modest quality raises the portfolio Sharpe more than a large improvement to the best single strategy would.

8.2 Allocating capital across survivors

The default allocation across low-correlation survivors is *equal risk*, not equal capital: weight each strategy inversely to its volatility so that each contributes roughly the same risk to the book, rather than letting the most volatile strategy dominate the drawdowns. Equal-risk weighting is robust precisely because it needs no return forecast — it uses only volatilities and, in its fuller form, correlations, which are estimated far more reliably than expected returns. Tilting away from equal risk toward strategies with higher measured Sharpe is defensible only to the degree those Sharpe estimates are trustworthy, and after the overfitting-control paper one holds them with suspicion; the honest default is to let diversification, not conviction, set the weights, and to cap any single strategy's risk contribution so that one strategy's death — which Section 7 promises will eventually come — cannot sink the book.

There is a deeper reason to resist the temptation to tilt hard toward the highest-Sharpe strategy, and it is estimation error. The full mean-variance-optimal weights — the ones that maximize the portfolio Sharpe if you knew the true means, volatilities, and correlations — are notoriously unstable: they place enormous bets on small differences in estimated expected returns, and since expected returns are the least reliably estimated quantity in all of finance, the "optimal" portfolio is usually a leveraged bet on estimation noise. Equal-risk weighting is deliberately *not* optimal under known inputs; it is optimal under the honest admission that you do *not* know them, and it degrades gracefully when your estimates are wrong because it never bet much on any of them. This is the portfolio-level echo of the paper's recurring theme: the robust choice is the one that assumes less. A book of low-correlation edges, weighted by risk rather than by forecast, held under total-heat and correlated-exposure caps, and pruned by a disciplined graveyard, is not the portfolio that would have performed best in the backtest — it is the portfolio most likely to survive the part of the future the backtest could not see.

8.3 When the meta-portfolio does not help

Honesty requires the boundary condition. Diversification delivers only what the correlations allow, and its two failure modes are worth pre-empting. First, *correlations rise in crises*: strategies that look independent in calm markets can become one position in a crash, so the equal-risk weights computed on quiet-period correlations understate joint risk exactly when it matters, which is why the correlated-exposure and total-heat caps of Section 3.4 sit above the portfolio weighting rather than being replaced by it. Second, *adding a bad strategy is not diversification*: a component with no real edge does not improve the portfolio by being uncorrelated; it adds noise and cost while contributing nothing, and dressing it up as "diversifying" is a common way to smuggle a dead strategy back onto the book. The meta-portfolio beats any single algorithm only when its components are *individually* real and *jointly* weakly correlated — both conditions, not either alone.

9 Reproducibility and the ethics of publishing your own results

This paper publishes an author's own trading results, which raises a governance problem sharper than any faced in a purely methodological piece: the same person designed the strategies, ran the analyses, and is now presenting the outcome, with every incentive to present it flatteringly. This closing section states the rules that keep such a

document honest — both the mechanical reproducibility that the backtesting and overfitting-control papers require and the specific ethical line that self-published trading results must not cross.

9.1 What to show and what to redact

The rule adopted throughout Section 6 is to publish *aggregate, risk-normalized* metrics and nothing else: profit factors, Sharpe ratios, hit rates, R-multiple distributions, drawdown shapes, and train-versus-test ratios. Absolute capital, position sizes in currency, per-contract costs, account values, and broker details are redacted not because they are secret but because they are *irrelevant to the claim and revealing of the person* — the edge is a property of the strategy, fully expressed in ratios and R-units, and the dollar figures add nothing to the reader's ability to evaluate it while exposing information that is no one's business. A stricter line governs licensed data: where a signal depended on a proprietary or licensed market-data feed, the values from that feed are *never* reproduced, and the dependency is disclosed as exactly that. These are not stylistic choices; they are the conditions under which publishing one's own results is legitimate at all. Every private number in this paper is labelled *private trade list, not redistributed*, and the case-study section carries an explicit publish-review flag for that reason.

It is worth being concrete about what this redaction removed from Section 6, because the abstraction is easy to nod along with and the practice is where it bites. The source material for those cases contained, for every strategy, the absolute net profit in currency, the account's starting capital, the per-trade risk budget and maximum contract count in currency and units, the per-contract commission-and-slippage assumption, the point value of the instrument, and the name of the trading platform and workspace the reports were exported from. *None of that appears here*. The profit factor 1.32 survived because it is a ratio; the net-profit figure it came from did not. The R-multiple distribution of Figure 4 survived because R normalizes away the currency; the risk budget it normalizes by was used inside the script and never printed. The point is that the redaction is not cosmetic — it discards the majority of what a raw backtest report contains — and that the surviving numbers are *sufficient*, on their own, for a reader to judge the strategy, which is the test of whether a redaction has removed the private without removing the informative. If the case for a strategy collapses once the dollar figures are gone, the dollar figures were doing persuasive work they should never have been asked to do.

9.2 Commit hashes and the append-only record

Reproducibility of the *general* results is mechanical and non-negotiable: every figure in Section 3, Section 5, and Section 8 is a seeded simulation whose script is named in its caption and whose numbers regenerate exactly from that script, and the case-study figure in Section 6 regenerates from the private list (which the reader cannot access, and which the caption says so). Behind the published numbers sits the discipline the backtesting paper mandated — each research note records the code path, the commit hash, the data version, the random seed, and the environment, and the results section is *append-only*: written after the run, never edited to fit a later story. The commit hash is the load-bearing detail. It timestamps *which* version of the code produced a number, which is the only defense against the most insidious form of self-deception in one's own research — quietly improving the code until the result improves, then presenting the improved result as though it had always been the plan. An append-only note with a commit hash makes that path visible; its absence makes it invisible.

9.3 The line between honest self-marketing and overfitting your own narrative

There is a legitimate version of publishing your own results — showing real work, real numbers, and real failures so that a reader can judge the method — and there is an illegitimate version that uses the apparatus of rigor to launder a curve-fit. The line between them is drawn by *what you do with the negative results*. The illegitimate

version reports the 1.44 profit factor and omits that it was the best of twenty variants; reports the winner and buries the graveyard; shows the equity curve and hides the drawdown bands that would let a reader see how ordinary it is; and narrates a clean story of insight where the truth was an accidental bug. The legitimate version — the one this paper has tried to be — does the opposite at every turn: it foregrounds the sign-flipped filter, the net-negative exit, the "leverage, not alpha" rejection, and the R-ladder that never fired; it labels the striking gate result *formulated, not validated*, and says exactly why; and it treats the reader as someone to be equipped rather than persuaded. The deepest form of overfitting is not to the data but to your own story about the data — selecting, from everything that happened, the subset that makes you look prescient. The only known defense is to pre-commit to reporting the whole record, winners and graveyard together, and to let the honest negative results do the work that a polished narrative cannot. A kill case, documented in full, is worth more than three winners' curves — because the kill is the one a reader cannot fake, and therefore the one they can believe.

Evidence basis

Every substantive claim in this paper rests on one of three kinds of basis, distinguished here so a reader can tell at a glance which is which — **own reproducible computation** (recomputed from free data or a self-contained model, reproducible via the cited script), **literature** (peer-reviewed article or textbook, cited inline), and **practitioner consensus** (desk practice with no clean citation, treated as the weakest). Nothing here is called *validated* on external sources alone; the stronger word requires an own out-of-sample computation.

Claim / result	Basis
Half Kelly keeps about 75% of full-Kelly growth while roughly halving the chance of an underwater path (full Kelly's median max drawdown is 89%)	Own reproducible computation — figures/fig_sizing_curves.py
For an ordinary +0.20R edge the worst drawdown over 400 trades runs ~15R median, ~27R at the 95th percentile, ~35R at the 99th	Own reproducible computation — figures/fig_dd_bands.py
A pre-set one-sided CUSUM monitor detects a decayed edge with a median lag of about 108 trades	Own reproducible computation — figures/fig_monitoring.py
Three weakly-correlated edges blend to a portfolio Sharpe of 1.37, above the best single leg's 1.00 (diversification ratio 1.52)	Own reproducible computation — figures/fig_meta_portfolio.py
Case A R-multiple distribution: 2,284 trades, 28.2% hit rate, +0.165R expectancy, 1.32 gross profit factor, with a thin right tail carrying the edge	Own reproducible computation — private trade list, not redistributed; OOS split
Case A VIX gate lifts gross profit factor 1.32 → 1.44 (out-of-sample 1.62), but was selected post-hoc across ~20 variants	Own reproducible computation — private trade list, not redistributed; OOS split
Case B: the daily result is uncorrelated with the VIX level (Spearman $\rho \approx -0.03$), and an elevated-VIX short filter flips sign out-of-sample	Own reproducible computation — private trade list, not redistributed; OOS split
The Kelly criterion gives the growth-optimal betting fraction $f^* = p - (1-p)/b$	Literature — (Kelly 1956; Thorp 2006)
Volatility targeting scales exposure inversely to a forecast of the instrument's volatility	Literature — (López de Prado 2018)
Point-in-time data and code-level look-ahead avoidance are prerequisites for an honest backtest	Literature — (López de Prado 2018)
Entry, exit, sizing, and risk are one specification whose components interact, not four independent choices	Practitioner consensus — not independently verified
Hard/soft kill-switches, pre-committed retirement criteria, and equal-risk weighting under total-heat caps are the robust operating defaults	Practitioner consensus — not independently verified

References

- Bailey, D. H., & López de Prado, M. (2014). The deflated Sharpe ratio: Correcting for selection bias, backtest overfitting, and non-normality. *Journal of Portfolio Management*, 40(5), 94–107. Open access via SSRN: <https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2460551>. — The formal correction, consistent with the overfitting-control paper, for the multiple-testing problem invoked throughout Section 2.3, Section 6.1, and Section 7.3: the significance of the best of N trials must be deflated by N .
- Chan, E. P. (2013). *Algorithmic Trading: Winning Strategies and Their Rationale*. Hoboken, NJ: Wiley. — A practitioner's treatment of the full pipeline from signal to live execution, including capital allocation and the Kelly framing of Section 3; a useful bridge between academic method and operational reality.

- Kelly, J. L. (1956). A new interpretation of information rate. *Bell System Technical Journal*, 35(4), 917–926. — The original derivation of the growth-optimal betting fraction underlying Section 3.1–Section 3.3; a foundational and freely available paper.
- López de Prado, M. (2018). *Advances in Financial Machine Learning*. Hoboken, NJ: Wiley. — The reference for point-in-time data, look-ahead avoidance, backtest overfitting, and volatility-based position sizing that anchors Section 3.1 and Section 4.
- López de Prado, M. (2020). *Machine Learning for Asset Managers*. Cambridge: Cambridge University Press (Elements in Quantitative Finance). — A compact treatment of the same author's program, including the dangers of selection under multiple testing; complements the overfitting-control paper and Section 9.
- Sinclair, E. (2020). *Positional Option Trading: An Advanced Guide*. Hoboken, NJ: Wiley. — A rigorous practitioner account of turning an edge into a sized, risk-managed position, including fractional-Kelly reasoning and the discipline of retiring strategies; directly relevant to Section 2, Section 3, and Section 7.
- Thorp, E. O. (2006). The Kelly criterion in blackjack, sports betting, and the stock market. In S. A. Zenios & W. T. Ziemba (Eds.), *Handbook of Asset and Liability Management* (Vol. 1, pp. 385–428). Amsterdam: North-Holland. A publicly available version is hosted on the author's own site: <<https://www.edwardthorp.com/wp-content/uploads/2016/11/TheKellyCriterion2007.pdf>>. — The practical case for betting a *fraction* of Kelly, and the growth-versus-drawdown trade-off quantified in Section 3.2.

All references above are publicly accessible: two are open-access documents with direct links (the deflated-Sharpe paper on SSRN and the Thorp chapter on the author's site), one is a freely available foundational journal article, and four are published books. No proprietary, licensed-data, course, or trading-academy material is cited, and no value from a licensed market-data feed is reproduced anywhere in this paper.